

HyMeKo Exploration Summary

Results, documents, research venues, and reachable potential

Codex summary for Dr. Csaba Hajdu

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Abstract

This document consolidates the broad repository exploration performed on 2026-06-30. The short conclusion is that HyMeKo is now best understood as three things at once: a mature signed-hypergraph infrastructure stack, a disciplined experimental lab for testing whether structure helps learning and control, and a paper/evidence production system. The cleanest near-term contribution is the machine-verified cross-view consistency kernel. The highest-upside research lines are topology/arity matching for control, multiplicative holonomy/transport, and the new Fiber-Spike-Rotor language model. The RL line is promising but not yet a clean “HSiKAN wins control” story.

1 Executive read

The strongest current claim is not that HSiKAN beats every baseline. The strongest claim is that HyMeKo provides one canonical signed-hypergraph source of truth, emits many concrete views, and machine-checks that invariants recovered from those emitted views agree.

That claim is backed by the cross-view consistency work:

- `reports/2026-06-28-cross-view-consistency.md`
- `reports/2026-06-29-acceptance-novelty-execution.md`
- `verification/cross_view_consistency/`

The research layer is more volatile but unusually healthy: negative results are recorded, confounds are named, and the best follow-ups are discriminator tests rather than “train larger” guesses.

2 What the repository contains

3 Mature infrastructure

The framework layer looks mature enough to support papers and demos. The architecture is documented in `architecture/README.md`; the workspace uses a `parser/IR/query/emit` separation; benchmark CSVs in `hymeko_bench/results/` show compilation and expansion measurements; and the cross-view harness drives the real CLI emitters rather than a toy emitter.

The most valuable engineering property is the hub-and-spoke model:

$$X_f(\epsilon_f(H)) = Q(H)$$

where an emitted concrete view is parsed back into a common invariant and checked against the IR-level query. This turns “views should agree” into a testable commuting-square claim.

Area	Observed state
Rust framework	Workspace crates for parser, core IR, query engine, transforms, HRE, emitters, WASM/editor, server, MCP, wire protocol, formats, monitoring, graph, P-graph, Clifford, and Python bindings.
Evidence system	Plans under <code>docs/plans/</code> ; acceptance records under <code>reports/</code> ; milestone indexes in <code>DONE.md</code> , <code>BACKLOG.md</code> , and <code>ROADMAP.md</code> .
Transforms and views	URDF, SDF, MJCF, DOT, Mermaid, ROS2 launch, SysML-style requirements views, Gazebo/MuJoCo-related emitters, torch-dataflow templates.
Learning side	Signed-link, HSiKAN/SignedKAN/Gomb, Soma/RicciStim vision, HyMeYOLO, RL/control, StructuralActor, and the new <code>hymeko_lm/</code> .
Artifacts	Benchmark CSVs, checkpoints, generated figures, GIFs, PDFs, LaTeX reports, and experiment JSON/JSONL logs.

4 Core results

Line	Current finding	Status
Cross-view IR	16 robotics fixtures across five views, plus systems-engineering requirements fixtures; drift-prevention demo; Z3/sympy proof support.	strong
Topology matching	Matched controller topology wins 8/8 supervised plant targets; dense complete graph is not a universal winner.	strong
Arity matching	Hypergraph lift hurts graph plants but wins decisively on hypergraph plants. Match controller arity to plant arity.	strong
Signed-link	Strict/leakage-audited line is valuable; rotor geometry itself was not load-bearing in AUROC ablations. Structural features and protocol discipline are the real contribution.	mixed
Vision/Soma	Mean-pool falsification was partly a readout artifact. Spatial pyramid/tree readout gives a scalable position-aware compromise.	mixed
FSR language model	TinyShakespeare PoC: FSR pre-norm beats matched small transformer on bpb; sparse top-k fixes the quadratic wall; sphere residual loses.	open
RL/control	Infrastructure is serious, BC works, PPO success collapses, and no clean HSiKAN-over-MLP control win is established yet.	open

5 Vision line

The vision story is no longer a simple failure or success. Earlier mean-pool Soma/RicciStim results underperformed conventional controls. The June 29–30 readout work changes the diagnosis: the bottleneck was often positional compression.

Key result from `reports/2026-06-29-soma-spatial-tree-readout.md`:

Readout	Parameters	Cluttered-MNIST accuracy
Mean-pool	2010	0.314
Attention	2027	0.466
Spatial tree	5227	0.537
Flatten	24890	0.617

The useful principle is that a flatten readout over a spatial field can be compressed into a multi-scale spatial-pyramid pool: roughly one fifth the parameters for roughly ninety percent of the accuracy, with scale and variable-anchor support. This is exportable to HyMeYOLO heads, RL vision, and positioned graph/point readouts.

6 FSR language model

The new `hymeko_lm/` package is a clean-slate language-model experiment. The Fiber-Spike-Rotor mixer replaces softmax attention with a causal, signed, rotor-transported, spike-gated gather. The channel mixer uses HSiKAN/CR or Chebyshev-CR style machinery.

Reported Phase 1 result in `reports/2026-06-29-fsr-lm-phase1.md`:

Model	Validation bpb	Throughput
Transformer control	2.570	35.1k tok/s
FSR sphere	2.770	11.8k tok/s
FSR pre-norm	2.479	11.2k tok/s

The honest interpretation is important. The positive result belongs to the FSR mixer plus HSiKAN channel mix with a standard pre-norm residual. The Gomb sphere residual does not currently earn its place on this task. The speed report `reports/2026-06-29-fsr-lm-speed.md` adds hard top-k sparsity: `spike_k=16` preserves or improves quality and makes long context feasible, but absolute throughput still trails fused transformer attention.

7 Holonomy

The holonomy-discriminator line is the sharpest recent scientific update. The initial plan was to test whether cycle holonomy is load-bearing. The current implementation in `hymeko_rl/holonomy_probe.py` distinguishes additive signed-walk aggregation from multiplicative transport.

Verification run performed during this summary session:

Arm	Accuracy	Parameters
Transport / rotor-like multiplicative reader	1.000	49
Additive signed-walk reader	0.4805	225
MLP	0.5913	1505
Linear confound guard	0.5054	13

This is a useful but sobering result. It supports the idea that pure cycle holonomy is multiplicative/transport-like, but it downgrades expectations that plain additive HSiKAN message passing will solve parity-style holonomy tasks.

8 RL and control

The RL part is promising, but it is not yet a clean “HSiKAN wins RL” story. It should be described as an experimental substrate plus a developing theory of when structure should matter.

What is built

- MuJoCo/Gym-style tasks: reaching, pick-and-place, Galambos/collaborative, quadruped/walker variants, sanity worlds.
- PPO, BC warm-starts, off-policy experiments, rendering, checkpoints, and policy builders for MLP, HSiKAN, SA-HSiKAN, and mixtures.
- Fast testbeds in `hymeko_rl/sanity_rl.py` and `hymeko_rl/sanity_worlds.py`: bandit, grid, hex, and collaborative.
- A pre-screen entry in `hymeko_rl/rl_prescreen.py`.

Current empirical state

From `reports/2026-06-29-rl-overnight.md`, BC is the reliable worker: pick-and-place after BC reaches useful lift/place success. PPO then improves scalar return while collapsing actual task success to near zero. That is a reward-alignment problem, not yet an architecture win.

Policy	After BC lift/place	After PPO lift/place
HSiKAN	0.875 / 0.625	0.000 / 0.000
MLP	1.000 / 0.500	0.125 / 0.000

The right near-term RL claim is therefore not performance superiority. It is:

HyMeKo provides a structural, declarative substrate for control experiments; the standing hypothesis is that structure helps when the controller topology/arity matches the plant and task invariant.

Main blockers

1. **PPO success collapse.** Scalar return can improve while success gets worse. The fix path is BC anchoring, success-aligned terminal rewards, and reporting task success separately from return.
2. **Launch-bound HSiKAN.** Vanilla HSiKAN is slow in rollout because the forward is many tiny tensor operations. SA-HSiKAN and structural collapse are the right directions.
3. **Wrong tasks can hide structure.** Linear/local tasks let MLP tie. Holonomy, contact cycles, gait, and matched topology are better discriminators.

Best RL moves

- Use the fast toy lab before every slow MuJoCo run.
- Lead demos with BC policies until PPO refinement stops degrading success.
- Use SA-HSiKAN for rollout-heavy tasks.
- Validate topology matching in closed loop, not only supervised structural proxies.
- Reserve rotor/transport backbones for cyclic or holonomy-heavy tasks.

9 Best venues

10 Recommended next sequence

1. **Ship the cross-view consistency paper first.** It is the cleanest evidence package and the strongest novelty claim.

Venue family	Best matching story
IEEE T-SMC / IEEE SMC	Canonical hypergraph IR, machine-verified cross-view consistency, robotics plus SysML witnesses, drift-prevention demo.
Formal methods / MDE	Consistency by convention versus machine-verified commuting diagram over real emitters.
Control / CPS	Topology and arity matching: the controller should match the plant’s structural coupling, not merely have more edges or parameters.
Graph ML / signed networks	Leakage-audited structural features, strict protocols, and inductive signed-link learning.
ML architecture workshop	FSR language model with rotor transport, spike top-k, and HSiKAN channel mix; must scale beyond TinyShakespeare before strong claims.
Vision / neuro / perception	Spatial-pyramid readout, RicciStim structure diagnostics, and possible brain-predictivity rather than raw accuracy claims.
Robotics collaboration	Declarative control scenarios, BC-anchor refinement, matched topology, affect/reward models, and Kato-style collaborative grasping.

2. **Turn topology/arity matching into a closed-loop control result.** This is the natural next high-value RL/control discriminator.
3. **Make multiplicative holonomy the gauge mechanism.** Do not expect additive HSiKAN alone to solve parity-like holonomy.
4. **Repair RL refinement.** BC anchor plus success-aligned reward must precede more architecture claims.
5. **Scale FSR carefully.** Lead with `spike_k=16`; treat Gomb sphere as currently negative on byte-LM.
6. **Extract and reuse SpatialPyramidPool.** It is a practical component with obvious export routes.

11 Verification performed for this summary

The first attempt used system Python and failed because `torch` and `pytest` were unavailable. Retrying through the project virtual environment succeeded:

```
.\.venv\Scripts\python.exe -m hymeko_rl.holonomy_probe --ring-size 12 --seeds 2
.\.venv\Scripts\python.exe -m pytest -q ^
hymeko_rl/tests/test_holonomy_probe.py hymeko_rl/tests/test_rl_prescreen.py
```

Result: transport solved the holonomy probe, additive/MLP/linear stayed near chance, and the two test files passed: **4 passed in 14.85 s.**

12 Final judgement

HyMeKo has one publishable formal-methods core now, several high-ceiling research programs, and a useful habit of recording falsification. The highest potential is not in claiming universal structural superiority. It is in showing exactly when structure is load-bearing: verified cross-view invariants, matched controller topology, matched arity, and multiplicative holonomy transport.